Riddhiman Raguraman *Software Engineer*

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Riddhiman Raguraman Riddhiman Raguraman Portfolio Website

Currently pursuing a Master's degree in Game Programming. Authorized to work in the United States under F-1 Optional Practical Training (OPT), with STEM extension eligibility.

Skills

C++
Optimization techniques, SIMD , Cache Hits, STL libraries, Multithreading Programming

Graphics and rendering
DirectX 11, HLSL, Compute Shaders

Other Computer Languages
C, C#, Java, Python, Assembly, and Bash.

Game Engines and Graphics Editors
Unreal Engine, Unity and Blender

Platforms
Windows, Linux

Build Systems and IDEs
Cmake, Premake, Visual Studio IDE

Projects

Game Engine Development , *(C++/DirectX11)*
- Built custom vector, matrix, and quaternion math libraries; achieved ~4× speedup using SIMD intrinsics (MMX, SSE).
- Implemented a high-performance file system using native OS APIs, including a custom archiving system for packed data storage.
- Designed a tree-based object hierarchy with scene graph and animation system support.
- Developed a DirectX-based rendering engine with vertex buffers, frustum culling, multiple cameras, object transforms, and mesh loading.
- Created an offline converter to transform GLTF meshes into a custom runtime format using Protocol Buffers.

Custom Malloc and Free , *(Windows C++ console project)*
- Designed and implemented a custom memory allocation and deallocation framework for performance-critical applications.
- Integrated tightly with C++ for low-level, system-level memory management.
- Achieved ~2× faster performance compared to MSVC's default allocator.

Multi threaded Maze solver , *(C++ multithreading Project)*
- Engineered a high-performance solver for massive 20K × 20K grids using bidirectional search.
- Launched concurrent threads from both start and end points to converge on a solution.
- Achieved 15–50% faster execution compared to traditional single-threaded BFS/DFS approaches.
- Implemented lock-free synchronization using C++ atomics to avoid mutex overhead.
- Utilized bitwise operations for efficient state representation and data processin

Multi threaded Audio engine , *(C++ Multithreading Project)*
- Engineered a real-time audio system to seamlessly stitch and play 23 raw WAVE files without artifacts.
- Designed a 25-thread architecture to decouple file I/O from audio rendering for glitch-free playback.
- Implemented a double-buffered Producer–Consumer model with circular queues for efficient inter-thread communication.
- Utilized mutexes and condition variables for safe thread synchronization (no spinlocks).
- Applied RAII for robust resource and thread lifecycle management under real-time constraints.

Cloud's Rescue , *(Unity 2d Project)*

- Developed an educational 2D mobile game combining story-driven gameplay with math-based challenges for 10th-grade students.
- Designed and implemented core game mechanics and dynamic level progression in C#.
- Integrated custom character assets created in Blender.
- Built using Unity 2D with ShaderLab for rendering and visual effects.

Professional Experience

05/2023 – 09/2023

Game Developer Intern, *Geonix Logix*
Worked on a 2D educational game called 'Clouds Rescue.' This game was designed to make learning math fun and interactive for 10th-grade students. Was responsible for developing the entire gameplay, handling level design, and writing all the interaction code.

Education

09/2024 – present
Chicago, Illinois

MS in Game Programming, *Depaul University*
CGPA: *2.92 / 4.00 (current)*

2020 – 2024
Chennai, India

Btech Computer Science, *VIT Chennai*
CGPA: *8.10 /10.00 | Equivalent 3.24 / 4.00*

Courses

06/2023 – 08/2024
Chennai, India

Master Certificate in Unreal AR/VR
(Monolith Research and Training labs)

03/2022
Chennai, India

Operating Systems and You: Becoming a Power User
(Coursera)

10/2022
Chennai , India

Data Science
(Internshala)

Academic Achievements

Game-A-Thon , *in VIT Chennai*

Secured first place in the 'Game-a-thon,' a 6-hour game development competition organized by Qubit as an integral component of the technical fest Technovit at VIT Chennai. The project involved the creation of a 2D platformer game, with interesting player mechanics.

IEEE YESIST12 2023 , *in NMIT Bengaluru*

My Project "Design of IOT-Based Agricultural Drones for Effective Crop Management - 5G and Beyond" was ***awarded as the BEST PROJECT*** in the preliminaries of the International Project Innovation Challenge for students and young professionals IEEE YESIST12 2023-'Kaushalya' Open House Project Expo-2023 Organized by Department of ECE, Nitte Meenakshi Institute of Technology, Bengaluru in association with IEEE Bangalore Section on 4" May 2023.

ICCSST - 2023 , *in Christ University Bengaluru*

Presented a conference paper on the topic "Social Distancing and Face Mask Detection Using YOLO Object Detection Algorithm" at the International Conference on Computational Sciences and Sustainable Technologies jointly conducted by CHRIST (Deemed to be University), Bangalore, India, and Modern College of Business and Science, Muscat, Oman. on May 8th, 2023, Our paper is yet to be published (as of July 31st 2023)

Languages

Tamil
Native Language

English
IELTS 7